EMRE ERTÜRK

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SUMMARY

Senior Computer Engineering student working on evolutionary computation, artificial life and agent-based simulation, moving toward bio-inspired robotics: evolutionary, developmental and swarm robotics. Co-author of a paper submitted to EMNLP 2026; TÜBİTAK 2209-A funded senior thesis.

EDUCATION

İZMIR INSTITUTE OF TECHNOLOGY **Urla, İzmir**

B.Sc. in Computer Engineering (GPA: 3.12; 3.83 in fall 2025–26; ALES: 85) **Sep 2022 – Feb 2027 (Expected)**

- Relevant courses: Behavioral Robotics (AA), Evolutionary Computation (BA), Embedded Computer Systems (BA), Analysis and Design of Algorithms (BA), Differential Equations (AA)

İZMIR ATATÜRK LISESI **İzmir**

High school; member of the Aviation and Space Club (İzmir Atatürk Lisesi Havacılık ve Uzay Kulübü, IALHUK)

RESEARCH

O. Kansoy, **E. Ertürk** (equal contribution). Smart Zebra: A Prompt-to-GRPO Recipe for Logic-Grid Reasoning in Small Language Models. *Submitted to EMNLP 2026.* Senior thesis, advised by Assoc. Prof. Selma Tekir, funded by TÜBİTAK 2209-A.

- SFT + GRPO training pipeline on Qwen2.5-3B-Instruct for logic-grid (Zebra) puzzles

E. Ertürk. AEASS v3: Revisiting an Adaptive Evolutionary Algorithm for Solving Sudokus. *Technical report, 2026.*

- Memetic algorithm (GA with min-conflicts local search); 65,201 of 65,201 runs solved over 52,458 benchmark puzzles, with a full ablation

E. Ertürk. Slavemaker Swarm: An Individual-Based Simulation of Polyergus Slave Raids on Formica Colonies. *Project report, 2026.*

- 3-D agent-based model with five pheromone fields, colony takeover and colony-level coevolution

PROJECTS

EVOMAXX (ARTIFICIAL LIFE SANDBOX) **2026 – Present**

- GPU artificial-life system in WebGL2 with generated music; controlled experiments on about 18,000 worlds raised the share of lifelike genomes from 7% to 95%
- Re-implemented twelve papers it builds on, including Lenia, Flow-Lenia, Physarum and MAP-Elites

P.C.B INDOOR DELIVERY ROBOT (CENG 483, TEAM OF FOUR) **Fall 2025**

- Arduino Mega and Jetson Nano robot with IMU-aided PID heading control, ultrasonic navigation and ArUco landmarks; three prototypes

PARTYBOT (CENG 424, TEAM OF FOUR) **Spring 2026**

- Gesture-controlled mecanum robot; sensor glove and robot linked by two ESP32 boards over ESP-NOW

EXPERIENCE

CODE 35 **İzmir**

Software Engineering Intern **Aug 2025**

- Built a passport OCR prototype (Tesseract, PassportEye) with MRZ-based error correction and check-digit validation
- Built a local Turkish customer-service chatbot with RAG on Ollama, compared small open models, and prototyped WhatsApp deployment
- Wrote feasibility studies and an interactive prototype for two new product ideas

TECHNICAL SKILLS

Programming: Python, C, JavaScript
Graphics & Simulation: WebGL2, GLSL, GPU compute, agent-based modelling
Robotics & Embedded: Arduino, ESP32, Jetson Nano, OpenCV
Methods: Evolutionary algorithms, local search, experimental design and statistical testing

LANGUAGES

Turkish (Native) • English (C1, YDS 93.5) • Chinese (Elementary) • German (Beginner)

LEADERSHIP & COMMUNITIES

IZTECH Nature and Mountain Sports Club (DDST) – President, Fall 2024

INTERESTS

Hiking, cycling, painting and reading